

Sales ML Project - Import Notes (Quick Revision)

1. pandas

```
import pandas as pd
```

Kyu Import Kiya?

- Dataset load karne ke liye
- DataFrame banane ke liye
- Data clean karne ke liye
- CSV read/save karne ke liye

2. numpy

```
import numpy as np
```

Kyu Import Kiya?

- Numerical operations
- Arrays
- Missing values (np.nan)
- Mathematical calculations

3. matplotlib.pyplot

```
import matplotlib.pyplot as plt
```

Kyu Import Kiya?

- Charts banane ke liye
- Accuracy comparison graph
- Model performance visualization

4. train_test_split

```
from sklearn.model_selection import train_test_split
```

Kyu Import Kiya?

- Dataset ko Training aur Testing data me divide karne ke liye

5. SimpleImputer

```
from sklearn.impute import SimpleImputer
```

Kyu Import Kiya?

- Missing values fill karne ke liye (Mean, Median, Most Frequent)

6. StandardScaler

```
from sklearn.preprocessing import StandardScaler
```

Kyu Import Kiya?

- Numeric features ko same scale par lane ke liye
- Mostly Logistic Regression, KNN aur SVM me

7. OneHotEncoder

```
from sklearn.preprocessing import OneHotEncoder
```

Kyu Import Kiya?

- Categorical data ko numeric format me convert karne ke liye (Male/Female → 0/1 etc.)

8. Pipeline

```
from sklearn.pipeline import Pipeline
```

Kyu Import Kiya?

- Preprocessing aur Model ko ek workflow me jodne ke liye

9. ColumnTransformer

```
from sklearn.compose import ColumnTransformer
```

Kyu Import Kiya?

- Numeric aur Categorical columns par alag preprocessing apply karne ke liye

10. LogisticRegression

```
from sklearn.linear_model import LogisticRegression
```

Kyu Import Kiya?

- Binary Classification Model banane ke liye

11. DecisionTreeClassifier

```
from sklearn.tree import DecisionTreeClassifier
```

Kyu Import Kiya?

- Decision Tree Classification ke liye

12. RandomForestClassifier

```
from sklearn.ensemble import RandomForestClassifier
```

Kyu Import Kiya?

- Multiple Decision Trees ka ensemble model banane ke liye

13. KNeighborsClassifier

```
from sklearn.neighbors import KNeighborsClassifier
```

Kyu Import Kiya?

- Nearest Neighbors ke basis par prediction karne ke liye

14. GaussianNB

```
from sklearn.naive_bayes import GaussianNB
```

Kyu Import Kiya?

- Probability-based Classification Model ke liye

15. SVC

```
from sklearn.svm import SVC
```

Kyu Import Kiya?

- Support Vector Machine Classification ke liye

16. accuracy_score

```
from sklearn.metrics import accuracy_score
```

Kyu Import Kiya?

- Model ki overall accuracy nikalne ke liye

17. confusion_matrix

```
from sklearn.metrics import confusion_matrix
```

Kyu Import Kiya?

- Correct aur Incorrect predictions dekhne ke liye

18. classification_report

```
from sklearn.metrics import classification_report
```

Kyu Import Kiya?

- Precision, Recall, F1-Score aur Support ek hi report me dekhne ke liye

19. precision_score

```
from sklearn.metrics import precision_score
```

Kyu Import Kiya?

- Positive predictions me kitni predictions sahi hain

20. recall_score

```
from sklearn.metrics import recall_score
```

Kyu Import Kiya?

- Actual positive cases me kitne detect hue

21. f1_score

```
from sklearn.metrics import f1_score
```

Kyu Import Kiya?

- Precision aur Recall ka balanced score nikalne ke liye

22. joblib

```
import joblib
```

Kyu Import Kiya?

- Trained model (.pkl) ko save aur load karne ke liye

Sales ML Project - Detailed Import Revision Notes

No	Import	Reason (Why Imported?)	Used In
1	pandas	Load, clean, analyze and save datasets using DataFrame	Entire project
2	numpy	Numerical calculations and NaN handling.	Preprocessing
3	matplotlib.pyplot	Create charts and graphs.	Visualization
4	train_test_split	Split dataset into training and testing sets.	Before training
5	SimpleImputer	Fill missing values so models can train.	Data cleaning
6	StandardScaler	Scale numerical features.	Logistic, KNN, SVM
7	OneHotEncoder	Convert categorical data into numbers.	Encoding
8	Pipeline	Create one complete ML workflow.	All models
9	ColumnTransformer	Apply different preprocessing to different columns.	Preprocessing
10	LogisticRegression	Train Logistic Regression model.	Model 1
11	DecisionTreeClassifier	Train Decision Tree model.	Model 2
12	RandomForestClassifier	Train Random Forest model.	Model 3
13	KNeighborsClassifier	Train KNN model.	Model 4
14	GaussianNB	Train Naive Bayes model.	Model 5
15	SVC	Train Support Vector Machine.	Model 6
16	accuracy_score	Calculate overall accuracy.	Evaluation
17	confusion_matrix	Check correct vs incorrect predictions.	Evaluation
18	classification_report	Get Precision, Recall, F1 together.	Evaluation
19	precision_score	Measure positive prediction correctness.	Evaluation
20	recall_score	Measure detected actual positives.	Evaluation
21	f1_score	Balance Precision and Recall.	Evaluation
22	joblib	Save and load trained .pkl model.	Deployment